

Department of Computer Science & Engineering@ JUET, Guna

Project Proposal format for students

B. Tech CSE (Batch2022-26) Major Project-2(18B19CI891)

Details of Student:
(IN BLOCK LETTERS)

Credits-4

S. No.	Names	Er. Numbers	Mobile No.	CGPA
1	Vaibhav Anand	221B426	8423859380	6.7
2	Uday Singh Parihar	221B417	9009834296	7.0
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Broad Title of project- Human Motion to Character Animation System

Area/Concept involved:

1. Computer Vision
2. Human Pose Estimation
3. Artificial Intelligence (Applied)
4. Computer Graphics & Animation
5. Motion Capture Systems

Language/Tools that may be used:

1. Programming Language: Python
2. Libraries & Frameworks: OpenCV, MediaPipe, NumPy
3. Animation Tool: Blender
4. Character Rigging: Mixamo
5. Interface :Streamlit

Prerequisite Knowledge Required-

Concept/ Subject	Software/ Tools
Basics of Computer Vision	Python programming
Fundamentals of Artificial Intelligence	OpenCV library
Linear Algebra (vectors, angles, transformations)	MediaPipe Pose framework
Human skeletal structure and kinematics	Blender (basic navigation and animation)
Basics of computer graphics and animation	JSON / file handling

Brief Description about project: (Minimum100 words)-

Motion capture is widely used in animation and gaming industries but often requires expensive hardware, sensor-based suits, and complex setups. This project aims to develop a low-cost, camera-based system that automatically extracts human motion from a normal video and applies it to a 3D animated character. The system uses computer vision and pose estimation techniques to detect human skeletal keypoints from video frames. These keypoints are processed, normalized, and converted into joint-based motion data. The extracted motion is then mapped onto a pre-rigged 3D character to generate realistic animation without the use of physical sensors or motion capture suits. The proposed system focuses on automation, affordability, and accessibility while maintaining acceptable motion

accuracy for animation purposes. This approach demonstrates how artificial intelligence and computer vision can simplify traditional animation pipelines.

Expected outcome at the end of 8th

- a. A functional system that converts a normal human video into character animation
- b. Successful extraction and visualization of human skeletal motion
- c. Generation of animation-ready motion data (BVH)
- d. A rigged 3D character animated using extracted human motion
- e. Demonstration of a complete AI-based automation pipeline
- f. Proper documentation, project report, and final demo

Signature of students

Consent of CSE dept Faculty (supervisor) -

Signature-

Name-

Consent of Other dept Faculty (if any) (supervisor) -

Signature-

Name-

HOD's Remark

Approved / Approved with changes / Not Approved

HOD's Signature

Date: